

# Physical Chemistry (I)

## Lecture 17



# 2차 중간고사 안내

- 시간: 5월 13일(목) 오후 1:30~2:30
- 장소: 화학관 315호(교수계획표에 나온 수업 장소)
- 계산기 불필요

# 시험 범위

- 5장 "헬름홀츠 및 깁스 에너지"
- 6장 "상 전이"
- 7장 "단순 혼합물" 진도 나간 곳까지

# 시험 문제

- 총 열 문제
- 과제 문제 혹은 그 변형
- 수업 시간에 배운 개념 설명
- 수업 시간에 배운 공식 유도
- 응용 문제

## Chapter 5

### Thermodynamic potentials

Clausius ineq.

First law

Internal energy

$$U$$

Helmholtz energy

$$A = U - TS$$

Enthalpy

$$H = U + pV$$

Gibbs energy

$$G = H - TS$$

Spontaneity

$$\Delta A_{T,V} \leq 0$$

$$\Delta G_{T,p} \leq 0$$

Maximum work

$$A_T = w_{\max}$$

$$G_{T,p} = w_{\text{add,max}}$$

Differential forms

Applications

- Gibbs-Helmholtz equation
- Maxwell relations
- Thermodynamic equation of state

Thermochemistry

Pressure dependence

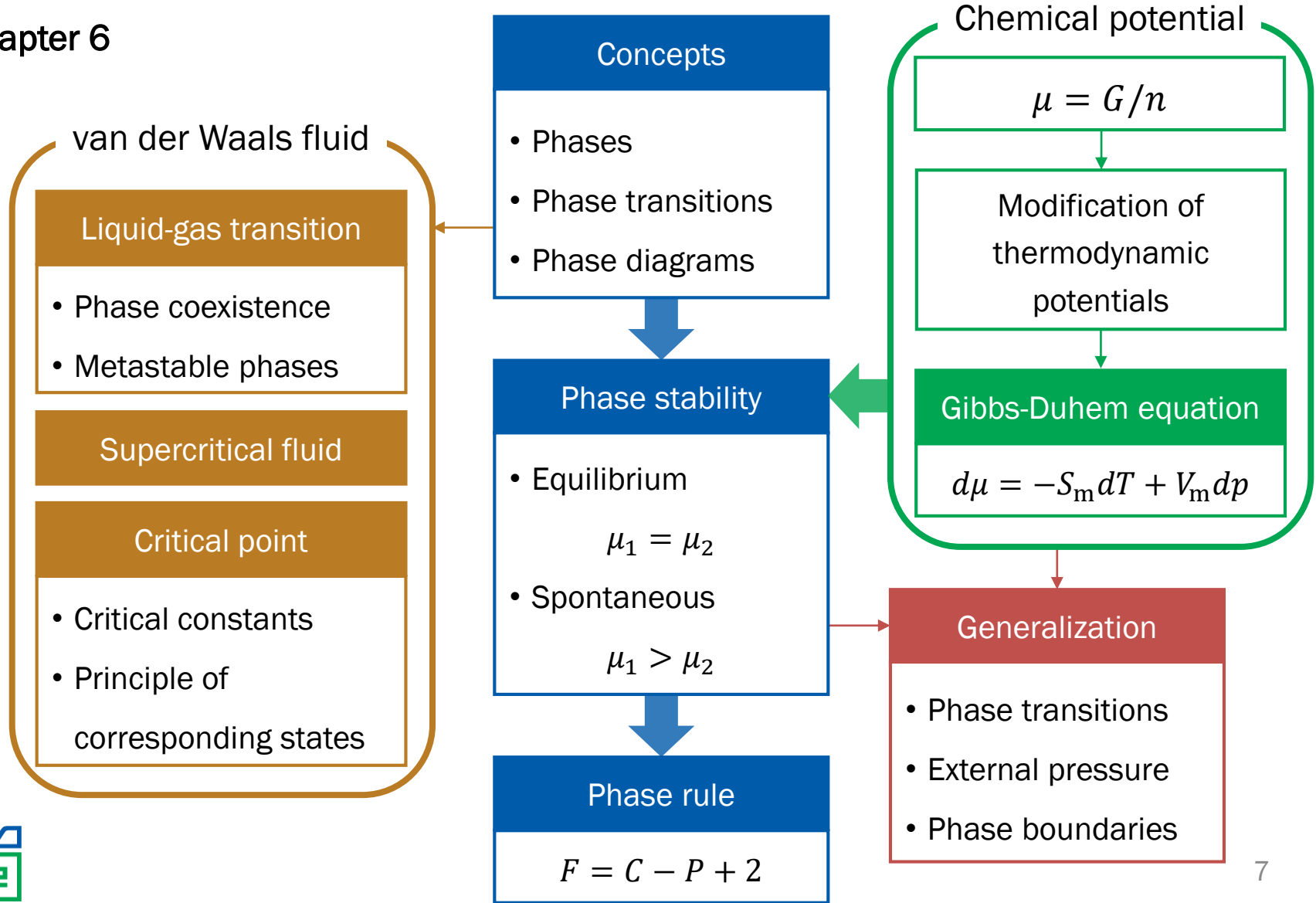


# Concepts and Derivations in Chpt 5

- [C] Helmholtz and Gibbs energies
- [D] Spontaneity: Clausius inequality
- [D] Meaning of Helmholtz and Gibbs energies
- [D] Thermodynamic potentials
- [D] Gibbs-Helmholtz equation
- [D] Maxwell relations
- [D] Thermodynamic equation of state
- [D] Pressure dependence of Gibbs energy



## Chapter 6



# Concepts and Derivations in Chpt 6

- [C] Phases and phase transitions
- [C, D] Chemical potential
- [D] Rewriting thermodynamic potentials
- [D] Gibbs-Duhem equation
- [D] Phase transition and chemical potential
- [C] Phase diagrams
- [C, D] Phase rule
- [C] Liquid-gas transitions





# Concepts and Derivations in Chpt 6

- [C] vdW: Liquid-gas transitions
  - [C] Maxwell construction
  - [D] Lever rule
  - [C] Binodal and spinodal lines
- [C] vdW: Supercritical fluid
- [C] vdW: Critical point
  - [D] Critical constants
  - [D] Critical compression factor
  - [D] Reduced variables
  - [C] Principle of corresponding states

# Concepts and Derivations in Chpt 6

- [C] Chemical potentials at transition
  - [D] Use of Gibbs-Duhem equation
- [D] Pressure effect on liquid-gas equilibrium
- [C] Phase boundaries
  - [D] Clausius equation
  - [D] Clausius-Clapeyron equation
  - [C] Interpretations

## Chapter 7

- Phase
- Chemical potential
- Gibbs-Duhem eq.

Multi-component system

### Gas mixture

Dalton's law

#### Perfect gas

- Chemical potential
$$\mu = \mu^\ominus + RT \ln(p/p^\ominus)$$
- $\Delta_{\text{mix}}G = -T\Delta_{\text{mix}}S$

#### Real gas

- Chemical potential
$$\mu = \mu^\ominus + RT \ln(f/p^\ominus)$$
- Fugacity  $f$
- $\Delta_{\text{mix}}G \neq -T\Delta_{\text{mix}}S$

### Liquid mixture

#### Ideal solution

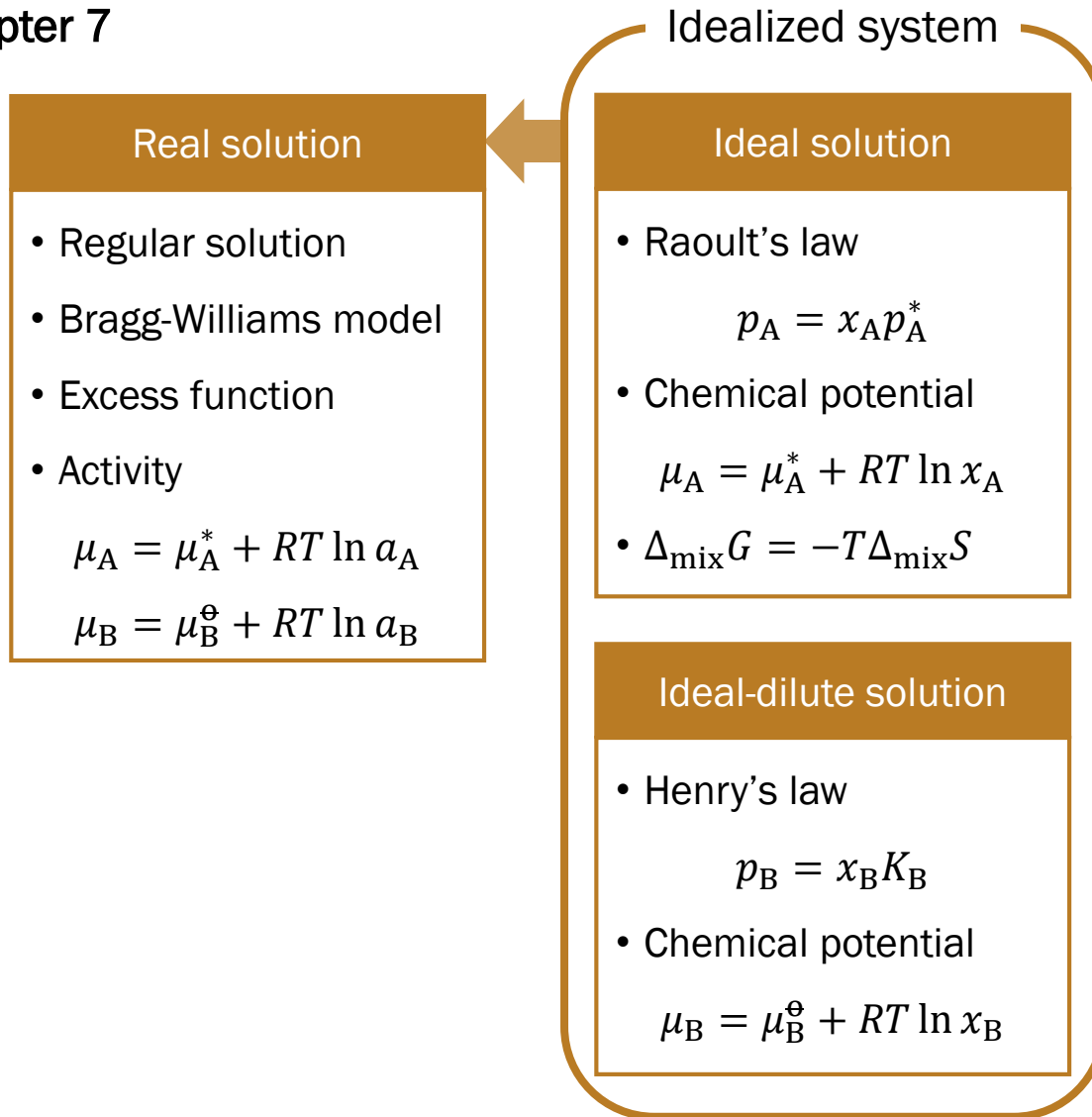
- Raoult's law
$$p_A = x_A p_A^*$$
- Chemical potential
$$\mu_A = \mu_A^* + RT \ln x_A$$
- $\Delta_{\text{mix}}G = -T\Delta_{\text{mix}}S$

#### Ideal-dilute solution

- Henry's law
$$p_B = x_B K_B$$
- Chemical potential
$$\mu_B = \mu_B^\ominus + RT \ln x_B$$

...

## Chapter 7



# Concepts and Derivations in Chpt 7

- Extension to multi-component systems
  - [C] Thermodynamic potentials
  - [C, D] Partial molar quantities
  - [C, D] Chemical potential
  - [D] Gibbs-Duhem equation
- [C] Dalton's law
- [D] Mixing of perfect gases
- [C] Fugacity and fugacity coefficient
- [C, D] Fugacity and pressure

# Concepts and Derivations in Chpt 7

- [D] Mixing of real gases
- [C] Mixing of liquids: liquid-gas equilibrium
- [C] Raoult's law and ideal solutions
- [D] Chemical potential of liquid
- [D] Mixing of ideal solution
- [D] Liquid and vapor compositions
- [C] Henry's law and ideal-dilute solutions
- [D] Connecting Raoult's and Henry's laws



# Concepts and Derivations in Chpt 7

- [C] (Idealized) chemical potentials of solute and solvent
- [C] Bragg-Williams model
  - [D] Entropy of mixing
  - [D] Energy of mixing
  - [D] Helmholtz energy of mixing
- [C] Regular solutions
- [C, D] Excess function
- [C, D] Activity and activity coefficient
  - [D] Activity of Bragg-Williams model